**DEPARTMENT OF MATHEMATICS AND APPLIED MATHEMATICS**

# Mathematics III Topics in Analysis (3TN)

**Tutorial Sheet 2: Solutions** **1 August 2024**

1. Let $(X,d)$ be a metric space. Let $E, F$ be two subsets of $X$. Show if $E$ is open and $E \cap F = \emptyset$, then $E \cap \overline{F} = \emptyset$, where $\overline{F}$ is the closure of the subset $F$.

   By contradiction, assume that $E \cap \overline{F} \neq \emptyset$. Let $x_0 \in E \cap \overline{F}$. Now,

$$\left\{ \begin{array}{l} x_0 \in E, \\ E \text{ open} \end{array} \right. \quad \Rightarrow \quad \exists r > 0 \text{ such that } B(x_0, r) \subset E\,.$$

   Since, $x_0 \in \overline{F}$, we get by the dfinition of the closure that $B(x_0, r) \cap F \neq \emptyset$. Thus, we obtain that

$$\emptyset \neq B(x_0, r) \cap F \subset \underbrace{E \cap F = \emptyset}_{\text{By assumption}} \quad \text{(Here is the contradiction!)}$$

2. **(Uniform convergence and integration)**

   (a) Let $(f_n)$ be a sequence of continuous functions from $[0,1]$ to $\mathbb{R}$.

   (i) Define what is meant by the statement that $(f_n)$ converges pointwise to $f$.
   The sequence $(f_n)$ is said to converge pointwise to the function $f$ if, for very $x \in [0,1]$, the sequence of real numbers $(f_n(x))$ converges to $f(x)$. That is,

$$\forall \varepsilon > 0 \;\; \forall x \in [0,1] \;\; \exists n_\varepsilon(x) \in \mathbb{N} \text{ such that } n \geq n_\varepsilon(x) \quad \Rightarrow \quad |f_n(x) - f(x)| < \varepsilon\,.$$

   (ii) Define what is meant by the statement that $(f_n)$ converges uniformly to $f$.
   The sequence $(f_n)$ is said to converge uniformly to the function $f$ if

$$\lim_{n\to\infty} \sup_{x\in[0,1]} |f_n(x) - f(x)| = 0\,.$$

   That is,

$$\forall \varepsilon > 0 \;\; \exists n_\varepsilon \in \mathbb{N} \text{ such that } n \geq n_\varepsilon \quad \Rightarrow \quad |f_n(x) - f(x)| < \varepsilon \quad \forall x \in [0,1]\,.$$

   (iii) Show that if $(f_n)$ converges uniformly to $f$ then $\lim_{n\to\infty} \int_0^1 f_n(t)dt = \int_0^1 f(t)dt$.
   Hint: $\left| \int_0^1 f_n(t)dt - \int_0^1 f(t)dt \right| \leq \int_0^1 |f_n(t) - f(t)|dt$.
   Using the hint, we find that

$$\begin{aligned} 0 \leq \left| \int_0^1 f_n(t)dt - \int_0^1 f(t)dt \right| &\leq \int_0^1 |f_n(t) - f(t)|dt \leq \int_0^1 \sup_{x\in[0,1]} |f_n(x) - f(x)|dt \\ &= \sup_{x\in[0,1]} |f_n(x) - f(x)| \to 0\,. \end{aligned}$$

   Thus, it follows by the squeeze theorem that $\lim_{n\to\infty} \int_0^1 f_n(t)dt = \int_0^1 f(t)dt$.

(b) Consider the sequence $(f_n) : [0,1] \to \mathbb{R}$ defined for every $x \in [0,1]$ and every $n \in \mathbb{N}^*$ by

$$f_n(x) := \begin{cases} n^2 x(1-nx) & \text{if } 0 \le x \le \dfrac{1}{n}\,, \\ 0 & \text{otherwise}\,. \end{cases}$$

(i) Study the pointwise convergence of $(f_n)$.

According to the definition of the sequence $(f_n)$, we need to distinguish the case $x = 0$ to the cases with $x > 0$. This is due to the fact for every $x \in (0,1]$ there exists a number $n(x) \in N$ such that $\dfrac{1}{n(x)} < x$ and hence, for every $n \ge n(x)$, if follows that $\dfrac{1}{n} \le \dfrac{1}{n(x)} < x$ and therefore $f_n(x) = 0$. This shows that, for every $x \in (0,1]$, $\lim_{n\to\infty} f_n(x) = 0$. On the other hand,

$$f_n(0) = 0 \;\; \forall n \quad \Rightarrow \quad \lim_{n\to\infty} f_n(0) = 0\,.$$

We can now conclude that the sequence $(f_n)$ converges pointwise to the function $f$ identically equal to 0.

(ii) Evaluate the integral $\displaystyle\int_0^1 f_n(x)\,dx$ and say whether the sequence $(f_n)$ converges uniformly.

Notice that

$$\begin{aligned} \int_0^1 f_n(x)\,dx &= \int_0^{\frac{1}{n}} f_n(x)dx + \int_0^{\frac{1}{n}} f_n(x)dx = \int_0^{\frac{1}{n}} n^2x(1-nx)dx + \int_{\frac{1}{n}}^1 0dx \\ &= -n\int_0^{\frac{1}{n}} \frac{d}{dx}\Big[\frac{1}{2}(1-nx)^2\Big]dx = -n\Big[\frac{1}{2}(1-nx)^2\Big]_0^{\frac{1}{n}} = \frac{n}{2} \end{aligned}$$

Hence,

$$\int_0^1 f_n(x)\,dx = \frac{n}{2} \to \infty \quad \Rightarrow \quad (f_n) \text{ does not converge uniformly to 0 by (a)(iii)}\,.$$

(iii) Study the uniform convergence of $(f_n)$ on the interval $[a,1]$ with $a > 0$.

Hence, we want to check whether the sequence $(f_n)$ converges uniformly to 0 on $[a,1]$. Just notice that, since $a > 0$ there exists a positive integer $n(a) > 0$ such that $\frac{1}{n(a)} < a$. So, let $n \ge n(a)$, we get for every $x \in [a,1]$ that

$$\frac{1}{n} \le \frac{1}{n(a)} < a \le x \quad \Rightarrow \quad f_n(x) = 0 \text{ (from the definition } (f_n))\,.$$

Hence,

$$\forall n \ge n(a), \;\; \sup_{x\in[a,1]} |f_n(x)| = 0 \quad \Rightarrow \quad ((f_n) \text{ converges uniformly to 0 on } [a,1]\,.$$

3. (a) Show that

$$\arctan(s+t) \le \arctan s + \arctan t \qquad \forall s,t \ge 0\,.$$

We say that the function arctan is subadditive on $[0,\infty)$.

One way to establish an inequality between two functions is to check that such an inequality holds for their derivatives and then use the comparison property for the integrals. So, we fix $s \ge 0$ and consider the functions

$$f_s(t) := \arctan(s+t) \qquad \text{and} \qquad g_s(t) := \arctan s + \arctan t$$

for which

$$f_s'(t) = \frac{1}{1+(s+t)^2} \qquad \text{and} \qquad g_s'(t) = \frac{1}{1+t^2}\,.$$

Now, since $s \ge 0$, we have

$$\begin{aligned}
\frac{1}{1+(s+\tau)^2} \le \frac{1}{1+\tau^2} \quad \forall \tau \ge 0 \;\; &\Rightarrow \;\; \int_0^t \frac{1}{1+(s+\tau)^2}d\tau \le \int_0^t \frac{1}{1+\tau^2}d\tau \\
&\Rightarrow \;\; \arctan(s+t) - \arctan s \le \arctan t \\
&\Rightarrow \;\; \arctan(s+t) \le \arctan s + \arctan t\,.
\end{aligned}$$

(b) Let $(X,d)$ be a metric space and let $d' : X \times X \to \mathbb{R}$ be defined by

$$d'(x,y) := \arctan(d(x,y)) \quad \forall x,y \in X\,.$$

(i) Show that $d'$ is also a distance on $X$.
- $d'(x,y) \ge 0 \quad \forall (x,y) \in X \times X$ and $d'(x,y) = 0 \Rightarrow \arctan(d(x,y)) = 0 \Rightarrow d(x,y) = 0 \Rightarrow x = y$.
- $d'(x,y) = \arctan(d(x,y)) = \arctan(d(y,x)) = d'(y,x)$ $\forall (x,y) \in X \times X$.
- Let $x,y,z \in X$. So,

$$\begin{aligned}
d'(x,y) \;&=\; \arctan(d(x,z)) \le \arctan(d(x,y)+d(y,z)) \\
&\qquad \text{(since } d(x,z) \le d(x,y)+d(y,z) \text{ and arctan is increasing)} \\
&\le\; \arctan(d(x,y)) + \arctan(d(y,z)) \qquad \text{(using (a))} \\
&=\; d'(x,y) + d'(y,z)\,.
\end{aligned}$$

Therefore, the mapping $d'$ is a distance on $X$.

(ii) Say whether the distances $d$ and $d'$ are topologically equivalent.
$d$ and $d'$ are topologically equivalent since

$$d(x_n,x) \to 0 \quad \Leftrightarrow \quad d'(x_n,x) \to 0 \qquad \forall (x_n) \subset X,\; \forall x \in X\,.$$

(iii) Are they equivalent? Explain clearly your answer.
Notice that $d'(x,y) \le \frac{\pi}{2}$ and hence, the metric space $(X,d')$ is bounded while $(X,d)$ is not necessary bounded. Therefore, the distances $d$ and $d'$ are not equivalent.

4. (a) Let $A$ be a non-empty subset of $\mathbb{R}$. We recall that the diameter of $A$ is

$$\mathrm{diam}(A) := \sup\{|x-y|\colon\ x,y \in A\}\,.$$

Show that

$$\mathrm{diam}(A) = \sup\ A - \inf\ A\,.$$

First of all, notice that if $A$ is unbounded then $\mathrm{diam}(A) = infty$ and $\sup A - \inf A = \infty$. Now, assume that $A$ is bounded. For every $s,t \in A$

$$\boxed{s \le \sup A \quad \text{and} \quad t \ge \inf A \quad \Rightarrow \quad s-t \le \sup A - \inf A}\,.$$

By swapping $s$ and $t$ we get

$$|s-t| \le \sup A - \inf A \quad \forall s,t \in A \quad \Rightarrow \quad \mathrm{diam}(A) \le \sup A - \inf A\,.$$

Now, in order to prove the reverse inequality

$$\mathrm{diam}(A) \ge \sup A - \inf A$$

we use the definitions of $\sup A$ and $\inf A$ as follows.

$$\forall \varepsilon > 0 \quad \exists x_\varepsilon \in A \quad \text{such that} \quad \boxed{\sup A < x_\varepsilon + \frac{\varepsilon}{2} \quad \text{and}}$$
$$\exists y_\varepsilon \in A \quad \text{such that} \quad \boxed{\inf A > y_\varepsilon - \frac{\varepsilon}{2}}\,.$$

So,

$$\sup A - \inf A < x_\varepsilon - y_\varepsilon + \varepsilon \le |x_\varepsilon - y_\varepsilon| + \varepsilon \le \mathrm{diam}(A) + \varepsilon\,.$$

Therefore, letting $\varepsilon \searrow 0^+$ we obtain that

$$\sup A - \inf A \le \mathrm{diam}(A)\,.$$

(b) Let $f : X \to \mathbb{R}$ be a bounded function. Deduce that

$$\sup_{x\in X} f(x) - \inf_{x\in X} f(x) = \sup\{|f(x)-f(y)|\colon\ x,y \in X\}\,. \qquad (*)$$

just take $A = f(X) = \{f(x)\colon\ x \in X\}$ and notice that

$$\sup A = \sup_{x\in X} f(x), \qquad \inf A = \inf_{x\in X} f(x), \qquad \mathrm{diam}(A) = \sup\{|f(x)-f(y)|\colon\ x,y\in X\}$$

and hence, $(*)$ follows from (a).

5. Consider the sequence of functions $(f_n)$ defined by

$$f_n(x) = \frac{nx}{nx+1} \quad \forall x \in [0,1], \quad \forall n \in \mathbb{N}\,.$$

(a) Study the pointwise convergence of the sequence $(f_n)$.
Notice that $f_n(0) = 0$ and that for every $x \in (0,1]$,

$$\lim_{n\to\infty} f_n(x) = \lim_{n\to\infty} \frac{nx}{n\left(x + \frac{1}{n}\right)} = \lim_{n\to\infty} \frac{x}{x + \frac{1}{n}} = \frac{x}{x} = 1\,.$$

Hence, the sequemce $(f_n)$ converges pointwise to the function $f : [0,1] \to \mathbb{R}$ defined by

$$f(x) := \begin{cases} 0 & \text{if } x = 0, \\ 1 & \text{if } x \in (0,1]\,. \end{cases}$$

(b) Does the sequence $(f_n)$ converge uniformly? Explain your answer.

**First solution:** The sequence $(f_n)$ of **continuous functions** converges pointwise to the **discontinuous function** $f$. Therefore, $(f_n)$ does not converge uniformly.

**Second solution:** You could also use the sequence $x_n = \dfrac{1}{n}$ and $f_n(x_n) = \dfrac{1}{2}$ and get that

$$\sup_{x\in[0,1]} |f_n(x) - f(x)| \geq |f(x_n) - f(x)| = \frac{1}{2} \quad \Rightarrow \quad \liminf_{n\to\infty} \sup_{x\in[0,1]} |f_n(x) - f(x)| \geq \frac{1}{2}\,.$$

Thus the sequence $(f_n)$ does not converge uniformly.

6. Consider the sequence $(f_n)$ defined by $f_n(x) := \sin\left(\sqrt{x + 16n^2\pi^2}\right)$ for every $x \in [0,\infty)$ and for every $n \in \mathbb{N}$.

   (a) Show that there exists $K > 0$ such that $|f_n(x) - f_n(y)| \leq K|x-y| \quad \forall x, y \geq 0, \quad \forall n \in \mathbb{N}\,.$
   Hint: You can use $|\sin a - \sin b| \leq |a-b| \quad \forall a, b \in \mathbb{R}$ and $\sqrt{s} - \sqrt{t} = \dfrac{s-t}{\sqrt{s}+\sqrt{t}} \quad \forall s, t > 0.$
   Then find a number $K$.

   (b) Show that the sequence $(f_n)$ converges pointwise to 0.
   Hint: $\left(1 + \dfrac{x}{16n^2\pi^2}\right)^{1/2} \approx 1 + \dfrac{x}{32n^2\pi^2}$ for $n$ large. (This comes from $\sqrt{1+t} \approx 1 + \dfrac{t}{2}$ for $|t|$ small.)

   (c) Consider the sequence of real numbers $x_n := \dfrac{\pi^2}{4} + 4n\pi^2$, use the hint in (b) to find an approximation of $f_n(x_n)$ and deduce that the sequence $(f_n)$ does not converge uniformly.

7. Let $(X, d_X)$ and $(Y, d_Y)$ be two metric spaces. Let $f : X \to Y$ be function. Show that

$$f \text{ continuous} \quad \Leftrightarrow \quad \forall E \subset X, \quad f(\overline{E}) \subset \overline{f(E)}\,.$$

$\boxed{\Rightarrow}$ Assume that $f$ is continuous. Let $E \subset X$ and let show that $f(\overline{E}) \subset \overline{f(E)}$. Let $y \in f(\overline{E})$. We want to show that $B_Y(y, \varepsilon) \cap f(E) \neq \emptyset$ for every $\varepsilon > 0$. Now,

$$y \in f(\overline{E}) \quad \Rightarrow \quad y = f(x_0) \text{ for some } x_0 \in \overline{E}$$

By the continuity of $f$ at $x_0$, we get that there exists $\delta_{x_0,\varepsilon} > 0$ such that

$$x \in B_X(x_0, \delta_{x_0,\varepsilon}) \quad \Rightarrow \quad f(x) \in B_Y(f(x_0), \varepsilon) = B_Y(y, \varepsilon)\,.$$

On the other hand, $B_X(x_0, \delta_{x_0,\varepsilon}) \cap E \neq \emptyset$, since $x_0 \in \overline{E}$. Hence,

$$\emptyset \neq f(B_X(x_0, \delta_{x_0,\varepsilon}) \cap E) \subset f(B_X(x_0, \delta_{x_0,\varepsilon})) \cap f(E) \subset B_Y(y, \varepsilon) \cap f(E)\,.$$

Therefore, we obtain that

$$\forall y \in f(\overline{E}) \quad \forall \varepsilon > 0 \quad B_Y(y, \varepsilon) \cap f(E) \neq \emptyset \quad \Rightarrow \quad y \in \overline{f(E)} \quad \Rightarrow \quad f(\overline{E}) \subset \overline{F(E)}\,.$$

$\boxed{\Leftarrow}$ Assume for $f(\overline{E}) \subset \overline{f(E)}$ for every $E \subset X$ and that $f$ is not continuous. Hence, there exists $x \in E$ and a sequence $(x_n) \subset X$ which converges to $x$ while $(f(x_n))$ does not converge to $f(x)$. That is, there exists $\delta > 0$ such that

$$\forall N \in \mathbb{N} \quad \exists n_N > N \;\text{ such that }\; d_Y(f(x_{n_N}, f(x) \geq \delta\,.$$

Up to passing to a subsequence, we can further assume that the whole sequence $(x_n)$ is such that $d_Y(f(x_n), f(x)) \geq \delta \; \forall n \in \mathbb{N}$ $(*)$. Now, we set

$$E := \{x_n\colon\, n \in \mathbb{N}\}\,.$$

Using $f(\overline{E}) \subset \overline{f(E)}$, we get that $f(x) \in \overline{f(E)}$. Hence,

$$\begin{aligned} B_Y(f(x), \delta) \cap f(E) \neq 0 \quad &\Rightarrow \quad \exists n_{\delta,x} \in \mathbb{N} \text{ such that } f(x_{n_{\delta,x}}) \in B_Y(f(x), \delta) \\ &\Rightarrow \quad d_Y(f(x_{n_{\delta,x}}, f(x)) \geq \delta \;\; \text{(which contradicts } (*)\text{)}\,. \end{aligned}$$